



CSE422: Artificial intelligence

Quiz 03 [Fall 2025]

Time: 20 minutes, Full Marks: 15

A

Name:		ID:		Section:	
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1. In CSE422-AI lab, we use Moss for catching plagiarism. Moss is **95%** effective in detecting plagiarism, i.e. if a student has plagiarized from a peer or using AI, Moss correctly detects it in **95%** cases. However, Moss also has a **3%** false positives, where a student has not actually plagiarized but Moss still reports they did. From previous semesters we know that **30%** students typically plagiarize in their lab assignments of CSE422.

Now, a "friend" of yours (not you obviously) has been flagged for plagiarism by Moss and has got a 0. You don't know whether they actually plagiarized, but you DO know probability theory. Based on this, do you think your friend has actually plagiarized, or is it a case of false-positive report from Moss? Explain with mathematical reasoning. [7]

D: Moss detects plag

E: Actually plagiarized

$$\begin{aligned} P(E|D) &= \frac{P(D|E) \cdot P(E)}{P(D|E) \cdot P(E) + P(D|\neg E) \cdot P(\neg E)} \\ &= \frac{0.95 \times 0.3}{0.95 \times 0.3 + 0.03 \times 0.7} \\ &= \cancel{0.9694} \quad 0.9314 \end{aligned}$$

$$P(D|E) = 0.95$$

$$P(D|\neg E) = 0.03$$

$$P(E) = 0.3$$

$$P(\neg E) = 1 - P(E) = 0.7$$

∴ friend actually plagiarized.

2. Consider the following historical data on what types of cat to buy from a store.

Colour	Age	Breed	Buy?
Orange	4	Siamese	Buy
Black	3	Siamese	Buy
White	1	Persian	Buy
Black	1	Persian	Not buy
Black	5	Siamese	Not buy
Orange	2	Siamese	Buy
Orange	3	Persian	Buy
Black	2	Siamese	Not buy
White	1	Persian	Buy
Orange	3	Siamese	Not buy

Determine whether you should buy an orange Persian cat of age ≤ 2 using Naive-Bayes classifier. [8]

$$\begin{aligned}
 P(\text{buy} | \text{orange, persian, age} \leq 2) &= \frac{P(\text{orange, persian, age} \leq 2 | \text{buy}) \cdot P(\text{buy})}{P(\text{orange, persian, age} \leq 2)} \\
 &= \frac{P(\text{orange} | \text{buy}) \cdot P(\text{persian} | \text{buy}) \cdot P(\text{age} \leq 2 | \text{buy})}{P(\text{orange, persian, age} \leq 2)} \cdot P(\text{buy}) \\
 &= \frac{\frac{3}{6} \cdot \frac{3}{6} \cdot \frac{3}{6}}{\frac{6}{10}} \times \frac{6}{10} = \frac{\frac{3}{40}}{\frac{6}{10}} = \frac{0.075}{0.6}
 \end{aligned}$$

$$\begin{aligned}
 P(\text{-buy} | \text{orange, persian, age} \leq 2) &= \frac{\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{2}{4}}{\frac{4}{10}} \times \frac{4}{10} \\
 &= \frac{\frac{1}{80}}{\frac{4}{10}} = \frac{0.0125}{0.4}
 \end{aligned}$$

Buy.



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B

Name:		ID:		Section:	
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1. In CSE422-AI lab, we use Moss for catching plagiarism. Our not-so-sophisticated peers from a different course use JPlag. JPlag is 75% effective in detecting plagiarism, i.e. if a student has plagiarized from a peer or using AI, JPlag correctly detects it in 75% cases. However, JPlag also has a 20% false positives, where a student has not actually plagiarized but JPlag still reports they did. From previous semesters we know that 15% students typically plagiarize in their lab assignments of that course.

Now, a "friend" of yours (not you obviously) has been flagged for plagiarism by JPlag and has got a 0. You don't know whether they actually plagiarized, but you DO know probability theory. Based on this, do you think your friend has actually plagiarized, or is it a case of a false-positive report from JPlag? Explain with mathematical reasoning. [7]

D: JPlag detects plag

E: Friend actually plagiarized

$$\begin{aligned} P(E|D) &= \frac{P(D|E) \cdot P(E)}{P(D|E) \cdot P(E) + P(D|\neg E) \cdot P(\neg E)} \\ &= \frac{0.75 \times 0.15}{0.75 \times 0.15 + 0.2 \times 0.85} \\ &= 0.3982 \end{aligned}$$

$$P(D|E) = 0.75$$

$$P(D|\neg E) = 0.2$$

$$P(E) = 0.15$$

$$\begin{aligned} P(\neg E) &= 1 - 0.15 \\ &= 0.85 \end{aligned}$$

Since $P(E|D) < 0.5$, friend probably did not plag.

2. Consider the following historical data on what types of cat to buy from a store.

Colour	Age	Breed	Buy?
Orange	4	Siamese	Buy
Black	3	Siamese	Buy
White	1	Persian	Buy
Black	1	Persian	Not buy
Black	5	Siamese	Not buy
Orange	2	Siamese	Buy
Orange	3	Persian	Buy
Black	2	Siamese	Not buy
White	1	Persian	Buy
Orange	3	Siamese	Not buy

Determine whether you should buy an **orange Persian cat of age >2** using **Naive-Bayes** classifier. [8]

$$\begin{aligned}
 P(\text{buy} | \text{orange, persian, age} > 2) &= \frac{P(\text{orange, persian, age} > 2 | \text{buy}) \cdot P(\text{buy})}{P(\text{orange, persian, age} > 2)} \\
 &= \frac{P(\text{orange} | \text{buy}) \cdot P(\text{persian} | \text{buy}) \cdot P(\text{age} > 2 | \text{buy})}{a} \cdot P(\text{buy}) \\
 &= \frac{\frac{3}{6} \cdot \frac{3}{6} \cdot \frac{3}{6}}{a} \times \frac{6}{10} = \frac{3}{40} = \frac{0.075}{a}
 \end{aligned}$$

$$\begin{aligned}
 P(\text{buy} | \text{orange, persian, age} > 2) &= \frac{\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{2}{4}}{a} \times \frac{4}{10} \\
 &= \frac{\frac{1}{80}}{a} = \frac{0.0125}{a}
 \end{aligned}$$

$\therefore$ Buy.